

English Lecture Comprehension for Native-Arabic students

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Abstract

Language barriers pose significant challenges for Arabic-speaking students pursuing higher education, hindering their comprehension of course material and impeding academic performance. To address this issue, we propose an AI-driven platform that mitigates language-related obstacles by capturing and transcribing lecturers' speech and subsequently converting it into succinct Arabic lecture notes. This platform aims to enhance students' understanding of academic content in their native language, thereby breaking down linguistic barriers and facilitating a smoother educational journey. Our research involved comprehensive surveys and interviews, which underscored the widespread language-related difficulties faced by students during higher education. Feedback collected from users of our platform indicated marked improvements in lecture content comprehension, reduced processing time for information assimilation, and an overall enhancement in academic performance. The platform's effectiveness not only fosters improved academic outcomes but also broadens academic course choices, fostering easier communication and engagement. This study underscores the significance of addressing language barriers through innovative AI solutions to promote inclusivity and proficiency in higher education settings.

1. Introduction

In today's interconnected world, language significantly shapes education, with English emerging as the primary global communication tool. This fosters collaboration and knowledge exchange but also presents unique challenges for non-native English-speaking students, particularly in academic contexts where English holds sway. Arabic-speaking students frequently encounter substantial language barriers, impeding their grasp of intricate subjects and constraining interaction with academic materials. Even the most skilled

students may struggle due to advanced vocabulary and intricate language usage, potentially compromising their academic success. Surveys conducted with 120 students aspiring for higher education indicated that 71 percent encountered language-related issues. Furthermore, interviews with top students revealed that 73 percent faced similar challenges due to the escalating complexity of academic vocabulary.

To address this, we propose an AI-powered solution that transforms spoken English lectures into succinct Arabic notes. This platform aims to bridge the divide between Arabic-speaking students and English lectures, creating a more inclusive educational environment. Through translated and summarized lecture notes, Arabic students access a more comprehensive learning experience, breaking down language barriers that have traditionally hindered academic excellence and collaboration.

Our objective is to alleviate language-related challenges and foster a more inclusive, productive learning setting where all students can actively engage, comprehend course material, and attain academic goals. Common difficulties, such as limited personal interaction due to large class sizes, time constraints, and lack of one-on-one engagement, often hinder students from obtaining essential information from educators. These issues lead to missed learning opportunities and hinder overall educational progress.

- In Improving the teaching quality with a smart-education system [[Glukhov and Vasetskaya\(2017\)](#)]The article explores the transition of society, education, and universities towards "smart" systems driven by the modern smart economy. It highlights the shift from traditional and e-learning to smart education, focusing on quality specialist training. Attributes of smart universities like interactive learning environments and personalized paths are discussed. The article suggests adapting education to individual capabilities through precise metrics and electronic portfolios. However, a limitation is that smart education might not effectively

tackle language-specific barriers for non-native speakers. Integrating quality management principles is recommended for ongoing knowledge enhancement.

- In Improving education in developing countries: Lessons from rigorous impact evaluations [[Ganimian and Murnane\(2016\)](#)] The article reviews 223 impact evaluations of educational initiatives in low- and middle-income countries, categorizing interventions based on their theory of action. Four lessons are derived: Reducing costs and expanding schooling boosts attendance and attainment, not consistently student achievement. Informing parents about school quality and parenting practices impacts actions and child achievement. More/better resources enhance student achievement when changing daily school experiences. Well-designed incentives improve teacher effort and student achievement, especially for low-skilled teachers. The article addresses past conflicting conclusions by different grouping, offering insights beyond meta-analyses. Limitations include drawing conclusions from existing evidence and potential bias based on selected studies.

2. Proposed Solution

Prior research has significantly enhanced the existing education system; however, language has remained an overlooked aspect. Language barriers have the potential to impact both students and educators, obstructing comprehensive understanding of vital knowledge for students and impeding the seamless transfer of irreplaceable knowledge from educators to the next generation.

Our proposed solution, titled "Modawen Teaching Assistant," presents a comprehensive approach to mitigating the language barrier prevalent among Arabic students pursuing higher education. This multifaceted initiative comprises three integral components:

- **Speech to Text Model:** This component harnesses advanced speech to text technology to seamlessly convert the spoken words of the lecturer into written text. By transcribing the spoken content into text, the language barrier is bypassed, enabling Arabic-speaking students to access the educational material in a textual format.
- **Summarization Model:** Employing sophisticated summarization techniques, this model condenses the transcribed text into a summarized and coherent version. This step is crucial in ensuring that students receive a comprehensive yet concise rendition of the lecture content, facilitating better understanding and retention.
- **Machine Translation:** this component plays a crucial role in overcoming language barriers and ensuring the

preservation of contextual nuances. This is particularly important due to the varying interpretations of words within different educational disciplines.

- **Web Interface:** To enhance accessibility and user-friendliness, a user-oriented web interface forms an integral part of our solution. This interface provides a convenient platform for students to interact with the Modawen Teaching Assistant. Through this interface, students can access the transcribed and summarized lecture notes, thereby streamlining their learning experience.

2.1. Speech to text

Our decision to utilize a fine-tuned version of OpenAI's Whisper, specifically calibrated for lecture speech, is rooted in an evaluation of available options, taking into account the specific details of our project. The attractiveness of this fine-tuned Whisper model lies in its impressive capability to accurately transcribe spoken lecture content, even when faced with diverse accents, languages, and conversational complexities. The comprehensive fine-tuning process that this Whisper model has undergone further increases its suitability for our project's objectives. By opting for this advanced and specialized version of Whisper, we are confident in its ability to accurately convert spoken lectures into written text, offering a reliable and precise tool for our intended application. Through the strategic utilization of this fine-tuned Whisper model, we anticipate delivering an effective and inclusive speech-to-text solution that enhances the educational experience for Arabic-speaking students by providing them with accessible and comprehensible lecture content. [[Radford et al.\(2023\)](#)Radford, Kim, Xu, Brockman, McLeavey, and Sutskever]

2.2. Summarization Model

A key component to the success of our project is the summarization model. This element plays a crucial part in shortening the material of transcribed lectures to clear summaries, enhancing students' comprehension. Given the importance of this component, we have realized the need for an accurate model to be used, specifically for abstractive summarization. As a result, we fine-tuned the pre-trained model PEGASUS [[Zhang et al.\(2020\)](#)Zhang, Zhao, Saleh, and Liu] which has received praise for its skill in producing excellent abstractive summaries. [[Nalini et al.\(2023\)](#)Nalini, Narayan, Sridharan, and Pradhan]

2.3. Machine Translation

In our search for a robust machine translation solution, we assessed various options, with a specific focus on the Arabic language. [[Alkhatib and Shaalan\(2018\)](#)] Among these, we explored the Microsoft Translation API, Microsoft's API stood out due to its accuracy and versatility,

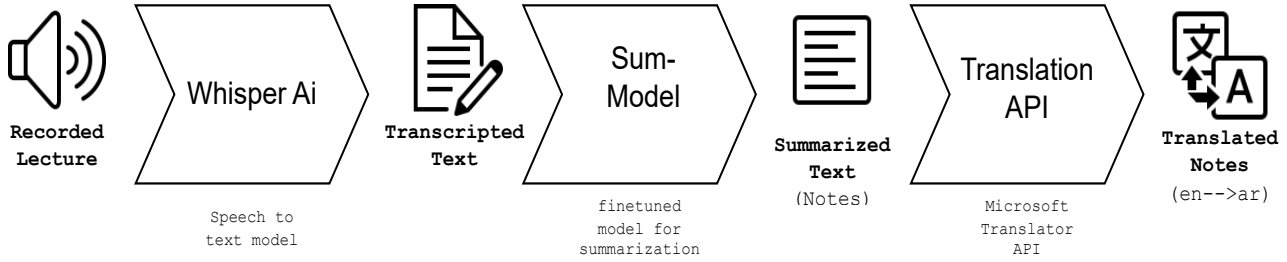


Figure 1. Framework Pipeline

making it well-suited for our objective of delivering clear and coherent translations for Arabic lecture content. [Almahasees(2021)] After conducting a comparative analysis between this API and other translation tools tailored for the Arabic language, we found that it captured the context very accurately, which will help our goal of creating a more inclusive and conducive learning environment, no matter the course of education.

2.4. Web Interface

With a deliberate emphasis on user-friendliness, our interface was meticulously designed to ensure effortless access to our features. Its intuitive layout enhance usability, providing a streamlined experience for users. By prioritizing accessibility and straightforward interactions, we aimed to create a platform that facilitates convenient engagement with our models.

3. Results & Discussions

Size	Parameters	Relative speed
tiny	39 M	~32x
base	74 M	~16x
small	244 M	~6x
medium	769 M	~2x
large	1550 M	1x

Table 1. Comparison of Model Sizes, Parameters, and Relative Speeds

In choosing our approach to the speech to text model, we found that training and building the whole model from scratch will cost us a lot more than simply using one that exists, openai’s Whisper, renowned for its capabilities, presented a range of speech-to-text models characterized by their varying sizes. 1 This classification based on size provided us with valuable options, each with its own merits and trade-offs. Our paramount consideration was the runtime latency, acknowledging its crucial impact on the user experience. With this perspective in mind, we strategically opted

for a fine-tuned iteration of the base model from the Whisper series. This selection represents a compromise between the number of model parameters and the relative speed at which the model processes incoming speech data.

By choosing the fine-tuned Whisper base model, we’re making the most of an existing solution and customizing it to match our exact needs for performance. This particular model setup strikes a great balance between quick processing and accurate results, creating a speech-to-text system that’s both versatile and fast. Our goal is to seamlessly integrate advanced tech into our platform, and this choice demonstrates our dedication to giving users a top-notch experience while being smart about how we use computing power and control expenses.

3.1. Datasets

When searching for available data, we found that the combination of two distinct datasets with different vocabulary levels will give results that match our needs during fine-tuning.

- The Augmented Multiparty Interaction Corpus dataset is a multi-modal dataset consisting of 100 hours of meeting recordings. The recordings use a range of signals synchronized to a common timeline. These include:

Close-talking and far-field microphones Individual and room-view video cameras Output from a slide projector and an electronic whiteboard Unsynchronized pens that record what is written during the meetings The meetings were recorded in English using three different rooms with different acoustic properties, and include mostly non-native speakers. The corpus is divided into two main parts:

Scenario meetings: These meetings were elicited using a scenario in which the participants play different roles in a design team, taking a design project from kick-off to completion over the course of a day. Naturally occurring meetings: These meetings were recorded in a variety of settings, including business meetings, aca-

demic seminars, and medical consultations. The AMI Meeting Corpus is a valuable resource for researchers in a variety of fields, including:

Speech and language processing Natural language understanding Multimodal signal processing Human-computer interaction Organizational psychology The corpus is freely available for download from the AMI website.

Here are some of the specific features of the AMI Meeting Corpus:

The recordings are synchronized to a common timeline, which allows for multimodal analysis of the data. The corpus includes a variety of meeting types, which makes it suitable for a wide range of research applications. The corpus includes recordings of both native and non-native speakers, which is important for research on cross-cultural communication. The corpus is well-annotated, which makes it easy to use for research purposes. The AMI Meeting Corpus is a valuable resource for researchers who are interested in studying human communication in meetings. It is a large and diverse corpus that can be used to address a wide range of research questions. [Renals and Swietojanski(2017)]

- The Scientific_Paper arXiv dataset is a collection of scientific papers that have been submitted to the arXiv eprint repository. The dataset is structured as a JSON file, with each line representing a scientific paper. The JSON file contains the following fields:

id: A unique identifier for the scientific paper. submitter: The name of the person who submitted the scientific paper. authors: A list of the authors of the scientific paper. title: The title of the scientific paper. comments: Additional information about the scientific paper, such as the number of pages and figures. journal-ref: Information about the journal the scientific paper was published in. doi: The Digital Object Identifier for the scientific paper. abstract: The abstract of the scientific paper. categories: A list of categories for the scientific paper. versions: A version history for the scientific paper. The dataset is divided into two parts: training and test. The training set contains 1.5 million scientific papers, and the test set contains 0.5 million scientific papers.

The Scientific_Paper arXiv dataset is a valuable resource for natural language processing. The dataset is large and diverse, and it provides a variety of features that can be used to train and evaluate Natural Language Processing models.

The id field is a unique identifier for the scientific paper. The submitter field contains the name of the per-

son who submitted the scientific paper. The authors field contains a list of the authors of the scientific paper. The title field contains the title of the scientific paper. The comments field contains additional information about the scientific paper, such as the number of pages and figures. The journal-ref field contains information about the journal the scientific paper was published in. The doi field contains the Digital Object Identifier for the scientific paper. The abstract field contains the abstract of the scientific paper. The categories field contains a list of categories for the scientific paper. The versions field contains a version history for the scientific paper.

Name	Accuracy	
	Train	Val
Whisper	0.86	0.71
Pegasus	0.89	0.83
Bert	0.94	0.991
Microsoft API	-	0.87

Table 2. Accuracy of Models

3.2. Fine-tuning PEGASUS

Throughout the process of fine-tuning our summarization model, we faced a tough challenge due to having limited computer power. To work around this, we came up with a smart plan to get the best out of our model despite these limits. We decided to focus on just two layers of both the encoder and decoder parts, leaving the rest frozen. This helped us balance out the need for speed and the ability to train our model.

As we dug deeper into making our model work better, we tried out different important things like how fast it learns, how much text it looks at once, and different ways to make sure it doesn't over-fit such as dropout. But even with all our hard work, we noticed that our model was getting too good at just the training data and not doing as well with new samples. So, we tried something new: we iterated between our two sets of data while training the model. This twist helped the model see a bigger picture and do better at generalizing to new samples. And found that the results were subpar, so we made a pivot to using BERT model, after fine-tuning bert it was repeating a lot in the summary so we had to do more iterations and using more data which resulted in a model that summarized well, with a faster runtime latency

Ibex played a crucial role in our progress by enabling us to cycle through the dataset and train on a larger amount of data at an accelerated pace. This efficiency not only sped up the fine-tuning process but also contributed to yielding superior outcomes within a shorter time-frame.

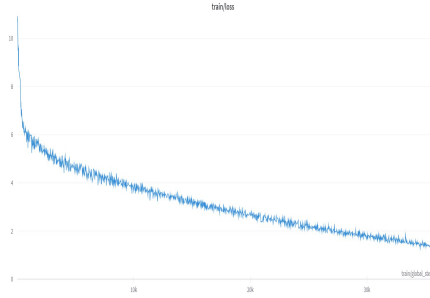


Figure 2. Training loss over time for PEGASUS

3.3. Limitations

As we navigate the landscape of advanced educational technologies, we confront a notable obstacle: the issue of runtime latency arising from the utilization of expansive speech-to-text and summarization models. Our foray into educational platforms involves integrating artificial intelligence to tackle tasks like transcribing spoken lectures and generating concise summaries. However, this shift amplifies the demand for computational resources and processing power. The resource-intensive nature of these functions can potentially introduce delays in real-time transcription and summarization, causing disruptions in the seamless dissemination of information and impeding swift access to translated content. It's worth acknowledging that the current scarcity of GPUs, pivotal for accelerating these computations, can further magnify the runtime latency challenge. Effectively tackling this dual predicament—balancing high computational requirements with limited hardware availability—assumes paramount importance as we endeavor to fully unlock the potential of these AI-driven solutions. Our priority is to ensure the effective and timely delivery of educational materials, particularly in dynamic classroom settings where immediate interaction and engagement are fundamental for achieving successful learning outcomes.

3.4. Future Work

Any work can be expanded, we have tried our best to implement this in the best way we see, but we would appreciate seeing expand beyond what we did.

- **Integration of Additional Languages:** One promising direction for future research is the inclusion of a broader range of languages in our speech-to-text and summarization models. This expansion would enable students who speak languages other than Arabic or English to equally benefit from our platform. By incorporating linguistic diversity, we can create a more inclusive learning environment that caters to students from various linguistic backgrounds.
- **Bidirectional Summarization:** Extending the summa-

rization feature to support translation and summarization from and to different languages is a crucial step towards achieving a truly multilingual platform. Enabling bidirectional summarization would facilitate the exchange of knowledge and ideas across language barriers, allowing students to access educational content in their preferred language while also promoting collaboration and cross-cultural understanding.

- **Enhancing Translation Accuracy:** Improving the accuracy and quality of translation between languages is paramount. Future efforts could involve refining our translation algorithms and leveraging advances in natural language processing (NLP) to ensure that the translated content maintains its original meaning, context, and nuance, thereby enhancing the learning experience for students using different languages.
- **User-Centric Design for Multilingual User Interface:** Designing a user interface that seamlessly accommodates multiple languages is crucial for providing an optimal experience to students worldwide. Future work should focus on developing a user-centric interface that allows users to easily switch between languages, access translated content, and engage with the platform's features effortlessly.
- **Cultural Sensitivity and Contextual Adaptation:** Globalizing the platform requires sensitivity to cultural nuances and educational contexts. Future research should explore how to adapt the content and user experience to align with different educational systems, cultural preferences, and local pedagogical approaches, ensuring that the platform is not only multilingual but also culturally relevant.
- **Accessibility and Technological Infrastructure:** To achieve widespread global impact, considerations must be given to the technological infrastructure and accessibility challenges faced by students in various regions. Future work should involve optimizing the platform's performance in regions with varying internet connectivity and device capabilities, ensuring that students from diverse backgrounds can equally access and benefit from the platform.

4. Conclusion

In summary, our Teacher Assistant program has brought significant improvements to our education system. By tackling the challenges of getting information and understanding due to language differences, we've created a better and more effective learning environment. Working together as a team, we carefully designed and smoothly integrated the Teacher Assistant. This new approach has been helpful in

providing customized help to students and making communication better between teachers and Arabic-speaking students. With this change we have made we try to lead to more active participation in class, less wasted time, and better academic results for the students we are aiming to help. Looking ahead, we believe this program will continue to push education forward. We want to thank everyone who played a part in making this project a success. Our strong commitment to always finding better ways to do things remains unchanged as we passionately work towards an educational system where every student can succeed

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